IMRS: Portable Innate Immune Response Score

Yansheng

2026-02-27

Contents

1 Phase II — Model Building (Anchors Only) 1

1.1 STEP 5A — Differential Expression (Anchors ONLY, Required) 1

1.2 STEP 6 — Anchor Reproducibility Filtering (Anchors ONLY) 1

1.3 STEP 6B — Anchor Heterogeneity / Variance Screening (Anchors ONLY) 1

1.4 STEP 6C — Meta-Level Power / Information Control (Anchors Only) 3

1.5 STEP 7 — Gene Weight Estimation (Anchors ONLY, Updated) 5

1 Phase II — Model Building (Anchors Only)

1.1 STEP 5A — Differential Expression (Anchors ONLY, Required)

Output:

```
results/<anchor_dataset_id>/de/delivery_vs_control.tsv
```

Columns:

- gene
- log2FC
- SE
- stat
- pval
- FDR

1.2 STEP 6 — Anchor Reproducibility Filtering (Anchors ONLY)

Output:

```
model/core_gene_set.tsv
```

1.3 STEP 6B — Anchor Heterogeneity / Variance Screening (Anchors ONLY)

Purpose: Detect genes whose delivery effect is unstable across anchors (platform/tissue sensitivity), and prevent them from dominating weights.

Input: Anchor DE tables for core genes only (log2FC + SE per anchor).

Process (per gene g)

Collect per-anchor effect estimates:

$$\beta_{g,i} = \text{log2FC}, \quad SE_{g,i}$$

Compute inverse-variance weights:

$$w_{g,i} = \frac{1}{SE_{g,i}^2}$$

Compute fixed-effect meta mean:

$$\hat{\beta}_g = \frac{\sum_i w_{g,i}\beta_{g,i}}{\sum_i w_{g,i}}$$

Compute heterogeneity (Cochran's Q):

$$Q_g = \sum_i w_{g,i} \left(\beta_{g,i} - \hat{\beta}_g\right)^2$$

with degrees of freedom:

$$df = K - 1$$

Let K_g denote the number of anchor datasets with non-missing $(\beta_{g,i}, SE_{g,i})$ for gene g.

Thus the degrees of freedom are defined per gene as:

$$df_g = K_g - 1$$

This value corresponds to the exported field `n_anchors_used`.

Convert to I^2 (bounded 0–1):

$$I_g^2 = \max\left(0, \frac{Q_g - df}{Q_g}\right)$$

Decision Rule (Pre-registered)

Flag "high heterogeneity'' if:

$$I^2 \geq 0.7$$

(I^2 is preferred because it is easier to interpret than Q.)

Action

Either:

- Drop flagged genes from weighting, or
- Penalize them during STEP 7 (recommended; less brittle)

Output

`model/gene_heterogeneity.tsv`

Columns:

- gene
- beta_meta
- se_meta
- Q
- I2
- n_anchors_used
- flag_high_heterogeneity

Interpretation: This is the variance check aligned with the principle:

$$\text{portability} > \text{performance}.$$

1.4 STEP 6C — Meta-Level Power / Information Control (Anchors Only)

Purpose

Although Step 6B controls cross-anchor heterogeneity using I^2, it does not control statistical uncertainty caused by insufficient effective sample size. Genes with large meta-level uncertainty (SE_{meta}) may introduce unstable weights and accumulate variance in the final score.

This step introduces a meta-level power screening procedure prior to weight estimation to control inferential reliability.

Inputs

- `model/core_gene_set.tsv`
- `model/gene_heterogeneity.tsv`

Required columns:

- gene
- beta_meta
- se_meta
- n_anchors_used

Statistical Framework

We approximate:

$$\hat{\beta}_g \sim \mathcal{N}(\beta_g, SE^2_{\text{meta},g})$$

Power Definition

For a pre-registered detectable effect size δ (default $\delta = 0.5$ log2FC) and two-sided significance level $\alpha = 0.05$:

$$z_\alpha = z_{1-\alpha/2}$$

Approximate statistical power:

$$\text{Power}_g \approx 1 - \Phi\left(z_\alpha - \frac{|\delta|}{SE_{\text{meta},g}}\right)$$

The above quantity represents detection power for a pre-specified biologically meaningful effect size δ using the meta-level standard error $SE_{\text{meta},g}$. It does not represent count-level power from the original DESeq2 model, but rather inferential strength of the meta-estimated effect.

Decision Rule

Flag "low-information'' if:

$$\text{Power}_g < 0.80$$

Optional safeguards:

- $n_{\text{anchors_used}} < 3$
- SE_{meta} in the worst quartile

Action Strategy

Apply a power penalty (preferred over hard exclusion):

$$\text{penalty}_{\text{power},g} = \begin{cases} 0.2 & \text{if Power}_g < 0.80 \\ 1.0 & \text{otherwise} \end{cases}$$

Output

`model/gene_power.tsv`

Columns:

- gene
- se_meta
- delta
- alpha
- power
- flag_low_power

Interpretation

This step controls uncertainty before weight formation and reduces propagation of noisy genes into the final score.

1.5 STEP 7 — Gene Weight Estimation (Anchors ONLY, Updated)

Purpose: Fix weights once, encoding magnitude + reliability.

Input:

- `model/core_gene_set.tsv`
- Anchor log2FC / SE
- `model/gene_heterogeneity.tsv` (Step 6B)
- `model/gene_power.tsv` (Step 6C)

Process

Compute inverse-variance meta effect:

$$\hat{\beta}_g = \frac{\sum_i w_{g,i}\beta_{g,i}}{\sum_i w_{g,i}}$$

Compute meta SE:

$$SE_{\text{meta},g} = \sqrt{\frac{1}{\sum_i w_{g,i}}}$$

Cap extremes to avoid domination:

$$\beta_g^{\text{capped}} = \text{clamp}(\beta_{\text{meta}}, [-c, +c])$$

where c is the 95th percentile of $|\beta_{\text{meta}}|$ across genes.

Heterogeneity penalty (Step 6B):

$$\text{penalty}_{\text{het},g} = 1 - I_g^2$$

Power penalty (Step 6C):

$$\text{penalty}_{\text{power},g} \in \{0.2, 1.0\}$$

Combined penalty:

$$\text{penalty}_{\text{combined},g} = \text{penalty}_{\text{het},g} \times \text{penalty}_{\text{power},g}$$

Optional stability floor:

$$\text{penalty}_{\text{final},g} = \max(\text{penalty}_{\text{combined},g}, 0.2)$$

Final weight:

$$w_g = \beta_g^{\text{capped}} \times \text{penalty}_{\text{final},g}$$

Output

`model/gene_weights.tsv`

Columns:

- gene
- weight
- beta_meta
- se_meta
- I2
- penalty_het
- penalty_power
- penalty_final
- capped_flag

Interpretation: Weights are fixed permanently and are never tuned using calibration or transfer datasets.